

Unit V: Project management

Estimation– FP Based

Function Point Analysis was initially developed by Allan J. Albrecht in 1979 at IBM and it has been further modified by the **International Function Point Users Group (IFPUG)**. The initial definition is given by **Allan J. Albrecht**.

Functional Point Analysis gives a dimensionless number defined in function points which we have found to be an effective relative measure of function value delivered to our customer.

Objectives of Functional Point Analysis

- The objective of FPA is to measure the functionality that the user requests and receives.
- The objective of FPA is to measure software development and maintenance independently of the technology used for implementation.
- It should be simple enough to minimize the overhead of the measurement process.
- It should be a consistent measure among various projects and organizations.

Types of Functional Point Analysis

There are basically two types of Functional Point Analysis, that are listed below.

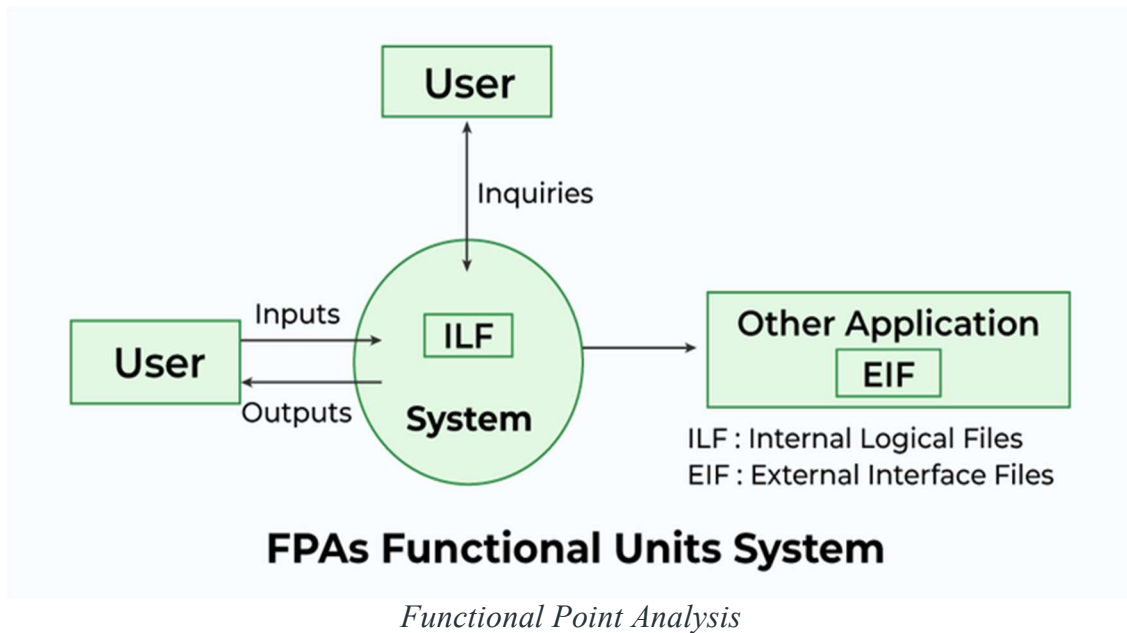
- Transactional Functional Type
- Data Functional Type

Transactional Functional Type

- **External Input (EI):** It is a elementary input parameter which is used in functional point analysis process where the input data's to the application are arranged from the external sources like input screen or business documents or external database or external tables or files i.e. the inputs to the application from outside but within business criteria boundary..
- **External Output (EO):** In this elementary process the processed data will be sent to the output in externally i.e. the generated response in terms of log reports or files. This report or files are used as an input for other application. The output screen or reports are examples of EO..
- **External Inquiries (EQ):** It is an elementary parameter, which provides relationship between input and output components to retrieve the response/output. The interrupts are the example of external inquiries.

Data Functional Type

- **Internal Logical File (ILF):** In this process the group of interrelated or logical data's are stored inside the system with related to the application standard. The input data's are processed by the application to stored in a file with logical structure. The database or directories are the examples of it.
- **External Interface File (EIF):** It is also known as external interface files where the group of logical interrelated external files are stored for reference purpose and used by the software application. It is externally maintained to validate/get the data from that file. The shared database or shared systems are the examples of it.



Benefits of Functional Point Analysis

- FPA is a tool to determine the size of a purchased application package by counting all the functions included in the package.
- It is a tool to help users discover the benefit of an application package to their organization by counting functions that specifically match their requirements.
- It is a tool to measure the units of a software product to support quality and productivity analysis.
- It is a vehicle to estimate the cost and resources required for software development and maintenance.
- It is a normalization factor for software comparison.

Characteristics of Functional Point Analysis

We calculate the functional point with the help of the number of functions and types of functions used in applications. These are classified into five types.

Measurement Parameters	Examples
Number of External Inputs (EI)	Input screen and tables
Number of External Output (EO)	Output screens and reports
Number of external inquiries (EQ)	Prompts and interrupts
Number of internal files (ILF)	Databases and directories
Number of external interfaces (EIF)	Shared databases and shared routines

Functional Point helps in describing system complexity and also shows project timelines. It is majorly used for business systems like information systems.

Weights of 5 Functional Point Attributes

Measurement Parameter	Low	Average	High
Number of external inputs (EI)	3	4	6
Number of external outputs (EO)	4	5	7
Number of external inquiries (EQ)	3	4	6
Number of internal files (ILF)	7	10	15
Number of External Interfaces (EIF)	5	7	10

Functional Complexities help us in finding the corresponding weights, which results in finding the Unadjusted Functional point (UFP) of the Subsystem. Consider the complexity as average for all cases. Below mentioned is the way how to compute FP.

Measurement Parameter	Count	Weighing Factor			
		Total_Count	Simple	Average	Complex
Number of external inputs (EI)	32	$32*4=128$	3	4	6
Number of external outputs (EO)	60	$60*5=300$	4	5	7
Number of external inquiries (EQ)	24	$24*4=96$	3	4	6
Number of internal files (ILF)	8	$8*10=80$	7	10	15
Number of external interfaces (EIF)	2	$2*7=14$	5	7	10
Algorithms used Count total →		618			

Example: Compute the function point, productivity, documentation, cost per function for the following data:

Measurement Parameter	Count		Weighing factor
1. Number of external inputs (EI)	24	*	4 = 96
2. Number of external outputs (EO)	46	*	4 = 184
3. Number of external inquiries (EQ)	8	*	6 = 48
4. Number of internal files (ILF)	4	*	10 = 40
5. Number of external interfaces (EIF) Count-total →	2	*	5=10 378

1. Number of user inputs = 24
2. Number of user outputs = 46
3. Number of inquiries = 8
4. Number of files = 4
5. Number of external interfaces = 2
6. Effort = 36.9 p-m
7. Technical documents = 265 pages
8. User documents = 122 pages
9. Cost = \$7744/ month

Various processing complexity factors are: 4, 1, 0, 3, 3, 5, 4, 4, 3, 3, 2, 2, 4, 5.

Solution:

So sum of all f_i ($i \leftarrow 1$ to 14) = $4 + 1 + 0 + 3 + 5 + 4 + 4 + 3 + 3 + 2 + 2 + 4 + 5 = 43$

$$\begin{aligned}
 FP &= \text{Count-total} * [0.65 + 0.01 * \sum(f_i)] \\
 &= 378 * [0.65 + 0.01 * 43] \\
 &= 378 * [0.65 + 0.43] \\
 &= 378 * 1.08 = 408
 \end{aligned}$$

$$\text{Productivity} = \frac{FP}{\text{Effort}} = \frac{408}{36.9} = 11.1$$

Total pages of documentation = technical document + user document
 $= 265 + 122 = 387$ pages

Documentation = Pages of documentation/FP
 $= 387/408 = 0.94$

$$\text{Cost per function} = \frac{\text{cost}}{\text{productivity}} = \frac{7744}{11.1} = \$700$$

Lines of Code (LOC) :

The **line of code** (LOC) metric is any line of text in a code that is not a comment or blank line, in any case of the number of statements or fragments of statements on the line. LOC consists of all lines containing program header files, declaration of any variable, and executable and non-executable statements. As Lines of Code (LOC) only counts the proportion of code, you can only utilize it to compare or estimate projects that utilize the same language and are programmed using the same coding standards.

So, now If LOC is simply a count of the number of lines then the above function shown contains 13 lines of code (LOC). But when comments and blank lines are ignored, the function shown above contains 12 lines of code (LOC).

Difference between LOC and Function Point :

Function Point (FP)	Line of Code (LOC)
Function Point metric is specification-based.	LOC metric is based on analogy.
Function Point metric is language independent.	LOC metric is dependent on language.
Function Point metric is user-oriented.	LOC metric is design-oriented.
Function Point metric is extendible to Line of Code.	It is changeable to FP (i.e. backfiring)
Function Point is used for data processing systems	LOC is used for calculating the size of the computer program
Function Point can be used to portray the project time	LOC is used for calculating and comparing the productivity of programmers.

In general, people prefer the functional size of software indicated as Function Point for one very important reason, i.e, the size expressed using the Function point metric stays constant in any case and whichever language or languages are used.

Features :

- Variations such as “source lines of code”, are used to set out a codebase.
- LOC is frequently used in some kinds of arguments.
- They are used in assessing a project’s performance or efficiency.

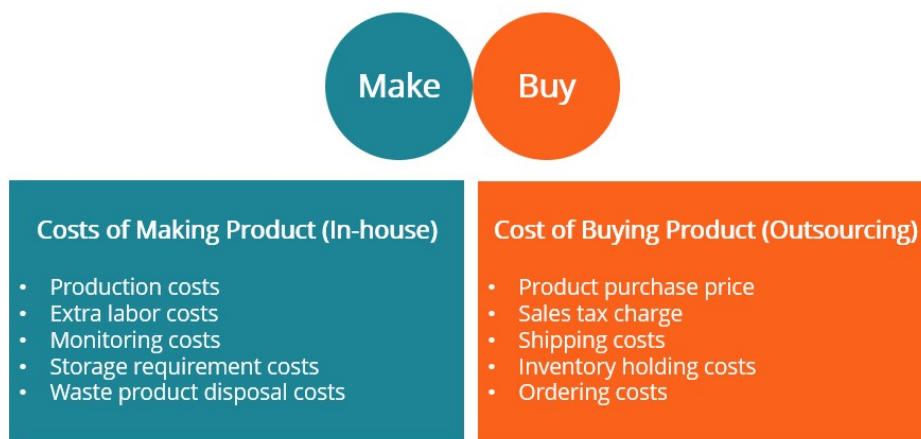
Advantages :

- Most used metric in cost estimation.
- Its alternates have many problems as compared to this metric.
- It is very easy in estimating the efforts.

Disadvantages :

- Very difficult to estimate the LOC of the final program from the problem specification.
- It correlates poorly with quality and efficiency of code.
- It doesn't consider complexity.

The Make or Buy Decision



Four Numbers You Should Know

When you are supposed to make a make-or-buy decision, there are four numbers you need to be aware of. Your decision will be based on the values of these four numbers. Let's have a look at the numbers now. They are quite self-explanatory.

- The volume
- The fixed cost of making
- Per-unit direct cost when making
- Per-unit cost when buying

Now, there are two formulas that use the above numbers. They are 'Cost to Buy' and 'Cost to Make'. The higher value loses and the decision maker can go ahead with the less costly solution.

Cost to Buy (CTB) = Volume x Per-unit cost when buying
 Cost to Make (CTM) = Fixed costs + (Per-unit direct cost x volume)

Reasons for Making

There are number of reasons a company would consider when it comes to making in-house. Following are a few:

- Cost concerns
- Desire to expand the manufacturing focus
- Need of direct control over the product
- Intellectual property concerns
- Quality control concerns
- Supplier unreliability
- Lack of competent suppliers
- Volume too small to get a supplier attracted
- Reduction of logistic costs (shipping etc.)
- To maintain a backup source
- Political and environment reasons
- Organizational pride

Reasons for Buying

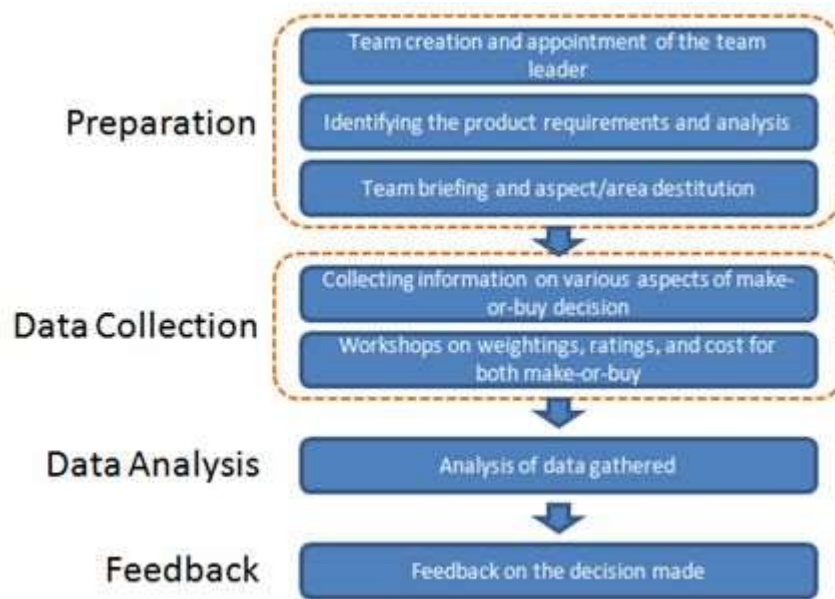
Following are some of the reasons companies may consider when it comes to buying from a supplier:

- Lack of technical experience
- Supplier's expertise on the technical areas and the domain
- Cost considerations
- Need of small volume
- Insufficient capacity to produce in-house
- Brand preferences
- Strategic partnerships

The Process

The make or buy decision can be in many scales. If the decision is small in nature and has less impact on the business, then even one person can make the decision. The person can consider the pros and cons between making and buying and finally arrive at a decision.

When it comes to larger and high impact decisions, usually organizations follow a standard method to arrive at a decision. This method can be divided into four main stages as below.



1. Preparation

- Team creation and appointment of the team leader
- Identifying the product requirements and analysis
- Team briefing and aspect/area destitution

2. Data Collection

- Collecting information on various aspects of make-or-buy decision
- Workshops on weightings, ratings, and cost for both make-or-buy

3. Data Analysis

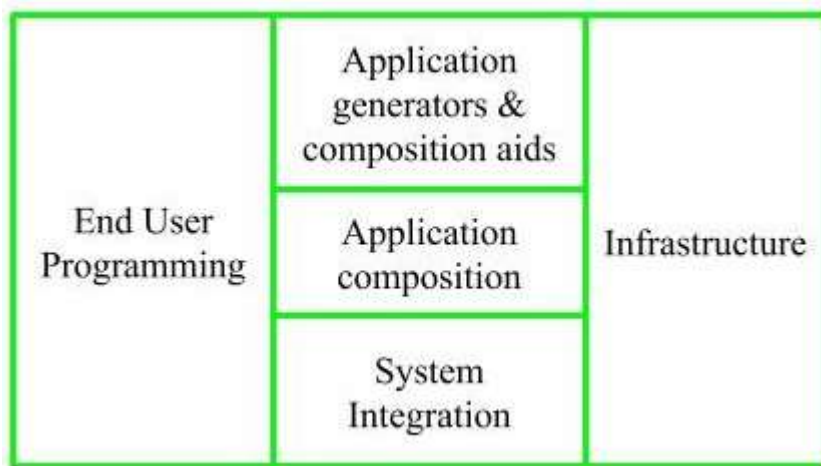
- Analysis of data gathered

4. Feedback

- Feedback on the decision made

COCOMO-II

COCOMO-II is the revised version of the original Cocomo (Constructive Cost Model) and is developed at University of Southern California. It is the model that allows one to estimate the cost, effort and schedule when planning a new software development activity. It consists of three sub-models:

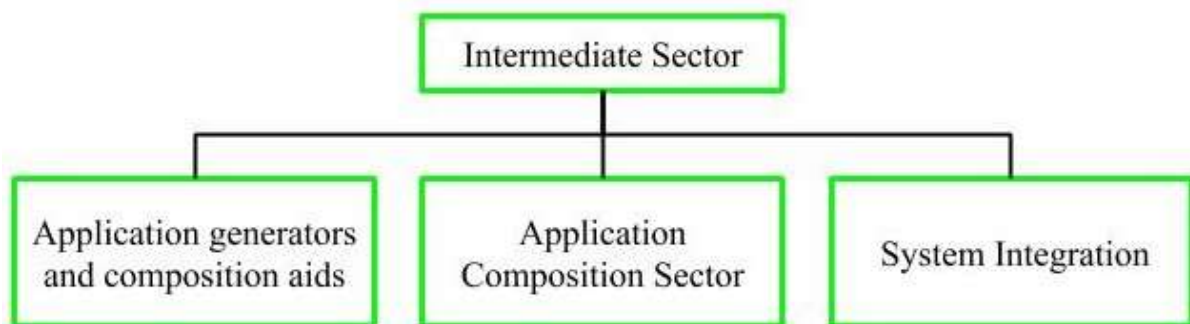


1. End User Programming:

Application generators are used in this sub-model. End user write the code by using these application generators.

Example – Spreadsheets, report generator, etc.

2. Intermediate Sector:



- **(a). Application Generators and Composition Aids –**
This category will create largely prepackaged capabilities for user programming. Their product will have many reusable components. Typical firms operating in this sector are Microsoft, Lotus, Oracle, IBM, Borland, Novell.
- **(b). Application Composition Sector –**
This category is too diversified and to be handled by prepackaged solutions. It includes GUI, Databases, domain specific components such as financial, medical or industrial process control packages.
- **(c). System Integration –**
This category deals with large scale and highly embedded systems.

3. Infrastructure Sector:

This category provides infrastructure for the software development like Operating System, Database Management System, User Interface Management System, Networking System, etc.

BASIC MODEL EQUATIONS

1. Effort = **a (KLOC)^b**. person month
2. Development Time = **c (Effort)^d** months

3. Number of People = **Effort / Development Time** persons

Software Product Type	a	b	c	D
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Product 1: A semi-detached mode product delivering 75,000 lines of code.

Product 2: An embedded mode product delivering 75,000 lines of code.

Product 3: An organic mode product delivering 75,000 lines of code.

Briefly comment on your estimations for the above products.

•

Solution:

We know that

Effort = **a (KLOC)^b**. person month

Development Time = **c (Effort)^d** months

Number of People = **Effort / Development Time** persons

Given that, estimated Lines of Code = 75,000 = 75 KLOC

Product 1: Semi-detached Mode of Development

Effort = **a (KLOC)^b**. person month

= **3 (75)^{1.12}** person month

= **377.73** person month

= **378** person month

Development Time = **c (Effort)^d** Months

= **2.5 (378)^{0.35}** Months

= **19.9** Months

= **20** Months

Number of People = **Effort / Development Time** persons

$$= 378 / 20 \text{ persons}$$

$$= 18.9 \text{ persons}$$

$$= 19 \text{ persons}$$

Product 2: Embedded Mode of Development

$$\text{Effort} = a (\text{KLOC})^b \text{ person month}$$

$$= 3.6 (75)^{1.2} \text{ person month}$$

$$= 640.288 \text{ person month}$$

$$= 640 \text{ person month}$$

$$\text{Development Time} = c (\text{Effort})^d \text{ Months}$$

$$= 2.5 (640)^{0.32} \text{ Months}$$

$$= 19.7 \text{ Months}$$

$$= 20 \text{ Months}$$

$$\text{Number of People} = \text{Effort} / \text{Development Time} \text{ persons}$$

$$= 640 / 20 \text{ persons}$$

$$= 32 \text{ persons}$$

Product 3: Organic Mode of Development

$$\text{Effort} = a (\text{KLOC})^b \text{ person month}$$

$$= 2.5 (75)^{1.05} \text{ person month}$$

$$= 232.67 \text{ person month}$$

$$= 233 \text{ person month}$$

$$\text{Development Time} = c (\text{Effort})^d \text{ Months}$$

$$= 2.5 (233)^{0.38} \text{ Months}$$

$$= 19.8 \text{ Months}$$

$$= 20 \text{ Months}$$

$$\text{Number of People} = \text{Effort} / \text{Development Time} \text{ persons}$$

= 233 / 20 persons

= 11.65 persons

= 12 persons

Project Planning

Once a project is found to be possible, computer code project managers undertake project designing. Project designing is undertaken and completed even before any development activity starts. Project designing consists of subsequent essential activities:



Estimating the subsequent attributes of the project:

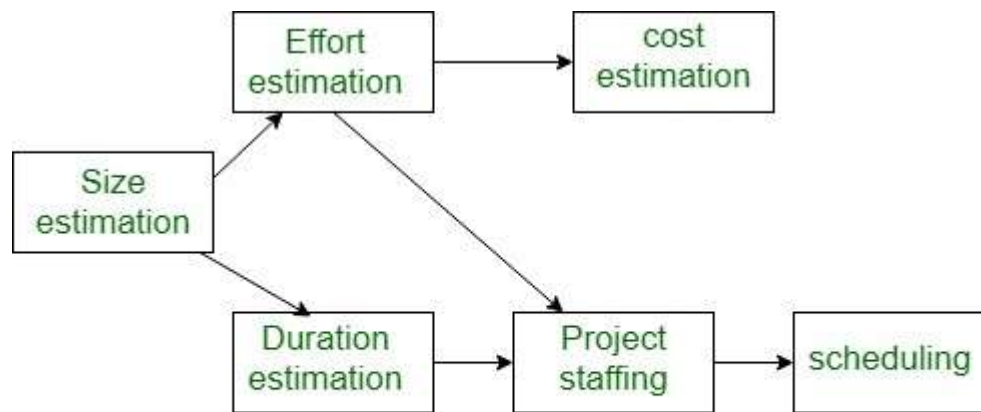
- **Project size:**
What's going to be downside quality in terms of the trouble and time needed to develop the product?
- **Cost:**
What proportion is it reaching to value to develop the project?
- **Duration:**
However long is it reaching to want complete development?
- **Effort:**
What proportion effort would be required?

The effectiveness of the following designing activities relies on the accuracy of those estimations.

- planning force and alternative resources
- workers organization and staffing plans
- Risk identification, analysis, and abatement designing
- Miscellaneous arranges like quality assurance plan, configuration, management arrange, etc.

Precedence ordering among project planning activities:

The different project connected estimates done by a project manager have already been mentioned. The below diagram shows the order during which vital project coming up with activities is also undertaken. It may be simply discovered that size estimation is that the 1st activity. It's conjointly the foremost basic parameter supported that all alternative coming up with activities square measure dispensed, alternative estimations like the estimation of effort, cost, resource, and project length also are vital elements of the project coming up with.



Precedence ordering among planning activities

Project planning includes the following 10 steps:

1. **Define stakeholders.** Stakeholders include anyone with an interest in the project. They can include the customer or end user, members of the project team, other people in the organization the project will affect and outside organizations or individuals with an interest.
2. **Define roles.** Each stakeholder's role should be clearly defined. Some people will fill multiple roles, however.
3. **Introduce stakeholders.** Hold a meeting to bring stakeholders together and unify the vision behind the project. The topics covered should include scope, goals, budget, schedule and roles.
4. **Set goals.** Take what is gleaned from the meeting and refine it into a project plan. It should include goals and deliverables that define what the product or service will result in.

5. **Prioritize tasks.** List tasks necessary to meet goals and prioritize them based on importance and interdependencies. A Gantt chart can be helpful for mapping project dependencies.
6. **Create a schedule.** Establish a timeline that considers the resources needed for all the tasks.
7. **Assess risks.** Identify project risks and develop strategies for mitigating them.
8. **Communicate.** Share the plan with all stakeholders and provide communications updates in the format and frequency stakeholders expect.
9. **Reassess.** As milestones are met, revisit the project plan and revise any areas that are not meeting expectations.
10. **Final evaluation.** Once the project is completed, performance should be evaluated to learn from the experience and identify areas to improve.



Through these steps, an organization can ensure that a project plan is reliable and well-communicated.

What are the 5 phases of a project?

Projects typically pass through five phases. The project lifecycle includes the following:

- **Initiation** defines project goals and objectives. It also is when feasibility is considered, along with how to measure project objectives.
- **Planning** sets out the project scope. It establishes what tasks need to get done and who will do them.
- **Execution** is when the deliverables are created. This is the longest phase of a project. During execution, the plan is set into motion and augmented, if necessary.
- **Monitoring and management** occur during the execution phase and may be considered part of the same step. This phase ensures that the project is going according to plan.
- **Closing and review** is the final Contracts are closed out and the final deliverables are given to the client. Successes and failures are evaluated.

RFP Risk Management– Identification, Projection

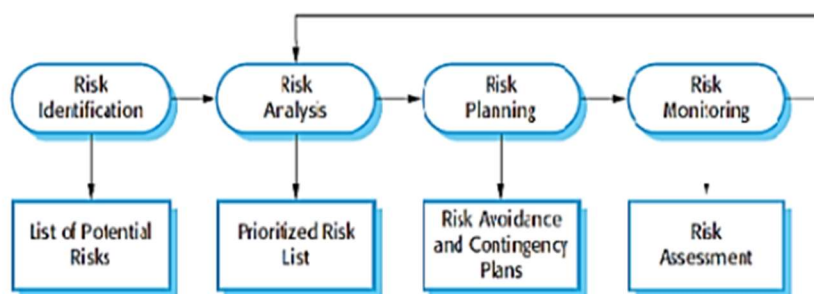


Fig 1. Risk management process

RFP (Request for Proposal) Risk Management in Software Engineering and Project Management involves the identification, assessment, and mitigation of potential risks associated with the process of responding to an RFP for a software engineering project. It specifically addresses the unique challenges and uncertainties that can arise during the proposal phase of a software development project. This process aims to minimize the adverse effects of risks on the successful completion of the project, ensuring that the proposal is both competitive and feasible within the defined constraints. RFP Risk Management in Software Engineering and Project Management involves:

Risk Identification:

- a. Create a risk register: Start by creating a comprehensive list of potential risks that could impact the project. Consider risks related to technology, resources, scope, time, and any other relevant factors.
- b. Involve stakeholders: Engage stakeholders, including the project team, clients, and subject matter experts, to ensure a comprehensive understanding of potential risks.
- c. Analyze historical data: Review past projects and industry data to identify common risks that may apply to your current RFP.

Risk Projection:

- a. Develop risk scenarios: Create possible scenarios for how identified risks could manifest during the project's lifecycle.
- b. Estimate risk impact: Quantify the potential impact of each risk scenario on the project's key parameters, including budget, timeline, and resource allocation.
- c. Assess risk tolerance: Determine the project's risk tolerance level and establish thresholds for acceptable risk levels based on the project's objectives and constraints.

Risk Mitigation Strategies:

- a. Develop a risk response plan: Formulate actionable strategies to mitigate, avoid, transfer, or accept each identified risk.
- b. Allocate resources: Allocate necessary resources, including time, budget, and personnel, to implement risk mitigation strategies effectively.
- c. Establish monitoring mechanisms: Implement a robust monitoring and control mechanism to track the effectiveness of risk mitigation strategies and promptly address any new risks that emerge during the project lifecycle.

Continuous Review and Update:

- a. Regularly review the risk management plan: Periodically reassess the project's risk landscape to incorporate new risks and make adjustments to the risk management plan as needed.

b. Foster a culture of risk awareness: Encourage proactive risk communication among team members and stakeholders to ensure everyone remains vigilant and responsive to potential risks throughout the project.

Remember, effective risk management is an ongoing process that requires continuous monitoring, evaluation, and adaptation to ensure the successful delivery of the project within the defined parameters.

Risk Mitigation, Monitoring, and Management (RMMM) plan

RMMM Plan:

A risk management technique is usually seen in the software Project plan. This can be divided into Risk Mitigation, Monitoring, and Management Plan (RMMM). In this plan, all works are done as part of risk analysis. As part of the overall project plan project manager generally uses this RMMM plan.

In some software teams, risk is documented with the help of a Risk Information Sheet (RIS). This RIS is controlled by using a database system for easier management of information i.e creation, priority ordering, searching, and other analysis. After documentation of RMMM and start of a project, risk mitigation and monitoring steps will start.

Risk Mitigation :

It is an activity used to avoid problems (Risk Avoidance).

Steps for mitigating the risks as follows.

1. Finding out the risk.
2. Removing causes that are the reason for risk creation.
3. Controlling the corresponding documents from time to time.
4. Conducting timely reviews to speed up the work.

Risk Monitoring :

It is an activity used for project tracking.

It has the following primary objectives as follows.

1. To check if predicted risks occur or not.
2. To ensure proper application of risk aversion steps defined for risk.
3. To collect data for future risk analysis.
4. To allocate what problems are caused by which risks throughout the project.

Risk Management and planning :

It assumes that the mitigation activity failed and the risk is a reality. This task is done by Project manager when risk becomes reality and causes severe problems. If the project

manager effectively uses project mitigation to remove risks successfully then it is easier to manage the risks. This shows that the response that will be taken for each risk by a manager. The main objective of the risk management plan is the risk register. This risk register describes and focuses on the predicted threats to a software project.

Example:

Let us understand RMMM with the help of an example of high staff turnover.

Risk Mitigation:

To mitigate this risk, project management must develop a strategy for reducing turnover. The possible steps to be taken are:

- Meet the current staff to determine causes for turnover (e.g., poor working conditions, low pay, competitive job market).
- Mitigate those causes that are under our control before the project starts.
- Once the project commences, assume turnover will occur and develop techniques to ensure continuity when people leave.
- Organize project teams so that information about each development activity is widely dispersed.
- Define documentation standards and establish mechanisms to ensure that documents are developed in a timely manner.
- Assign a backup staff member for every critical technologist.

Risk Monitoring:

As the project proceeds, risk monitoring activities commence. The project manager monitors factors that may provide an indication of whether the risk is becoming more or less likely. In the case of high staff turnover, the following factors can be monitored:

- General attitude of team members based on project pressures.
- Interpersonal relationships among team members.
- Potential problems with compensation and benefits.
- The availability of jobs within the company and outside it.

Risk Management:

Risk management and contingency planning assumes that mitigation efforts have failed and that the risk has become a reality. Continuing the example, the project is well underway, and a number of people announce that they will be leaving. If the mitigation strategy has been followed, backup is available, information is documented, and knowledge has been dispersed across the team. In addition, the project manager may temporarily refocus resources (and readjust the project schedule) to those functions that are fully staffed, enabling newcomers who must be added to the team to “get up to the speed”.

Drawbacks of RMMM:

- It incurs additional project costs.
- It takes additional time.
- For larger projects, implementing an RMMM may itself turn out to be another tedious project.
- RMMM does not guarantee a risk-free project, in fact, risks may also come up after the project is delivered.

Risk Management Process

The risk management process is a framework for the actions that need to be taken.

There are five basic steps that are taken to manage risk; these steps are referred to as the risk management process. It begins with identifying risks, goes on to analyze risks, then the risk is prioritized, a solution is implemented, and finally, the risk is monitored. In manual systems, each step involves a lot of documentation and administration.

Five Essential Steps of A Risk Management Process

1. Identify the Risk
2. Analyze the Risk
3. Evaluate or Rank the Risk
4. Treat the Risk
5. Monitor and Review the Risk

Step 1: Identify the Risk

The initial step in the risk management process is to identify the risks that the business is exposed to in its operating environment.

There are many different types of risks:

- Legal risks
- Environmental risks
- Market risks
- Regulatory risks etc.

It is important to identify as many of these risk factors as possible. In a manual environment, these risks are noted down manually. If the organization has a risk management solution employed all this information is inserted directly into the system.

The advantage of this approach is that these risks are now visible to every stakeholder in the organization with access to the system. Instead of this vital information being locked away in a report which has to be requested via email, anyone who wants to see which risks have been identified can access the information in the risk management system.

Learn in depth about [risk identification](#)



Step 2: Analyze the Risk

Once a risk has been identified it needs to be analyzed. The scope of the risk must be determined. It is also important to understand the link between the risk and different factors within the organization. To determine the severity and seriousness of the risk it is necessary to see how many business functions the risk affects. There are risks that can bring the whole business to a standstill if actualized, while there are risks that will only be minor inconveniences in the analysis.

In a manual risk management environment, this analysis must be done manually. When a risk management solution is implemented one of the most important basic steps is to map risks to different documents, policies, procedures, and business processes.

Step 3: Evaluate the Risk or Risk Assessment

Risks need to be ranked and prioritized. Most risk management solutions have different categories of risks, depending on the severity of the risk. A risk that may cause some inconvenience is rated lowly, risks that can result in catastrophic loss are rated the highest. It is important to rank risks because it allows the organization to gain a holistic view of the risk exposure of the whole organization. The business may be vulnerable to several low-level risks, but it may not require upper management intervention. On the

other hand, just one of the highest-rated risks is enough to require immediate intervention.

There are two types of risk assessments: Qualitative Risk Assessment and Quantitative Risk Assessment.

Qualitative Risk Assessment

Risk assessments are inherently qualitative – while we can derive metrics from the risks, most risks are not quantifiable. For instance, the risk of climate change that many businesses are now focusing on cannot be quantified as a whole, only different aspects of it can be quantified. There needs to be a way to perform qualitative risk assessments while still ensuring objectivity and standardization in the assessments throughout the enterprise.

Quantitative Risk Assessment

Finance related risks are best assessed through quantitative risk assessments. Such risk assessments are so common in the financial sector because the sector primarily deals in numbers – whether that number is the money, the metrics, the interest rates, or any other data point that is critical for risk assessments in the financial sector. Quantitative risk assessments are easier to automate than qualitative risk assessments and are generally considered more objective.

Step 4: Treat the Risk

Every risk needs to be eliminated or contained as much as possible. This is done by connecting with the experts of the field to which the risk belongs. In a manual environment, this entails contacting each and every stakeholder and then setting up meetings so everyone can talk and discuss the issues. The problem is that the discussion is broken into many different email threads, across different documents and spreadsheets, and many different phone calls. In a risk management solution, all the relevant stakeholders can be sent notifications from within the system. The discussion regarding the risk and its possible solution can take place from within the system. Upper management can also keep a close eye on the solutions being suggested and the progress being made within the system. Instead of everyone contacting each other to get updates, everyone can get updates directly from within the risk management

solution.

Step 5: Monitor and Review the Risk

Not all risks can be eliminated – some risks are always present. Market risks and environmental risks are just two examples of risks that always need to be monitored. Under manual systems monitoring happens through diligent employees. These professionals must make sure that they keep a close watch on all risk factors. Under a digital environment, the risk management system monitors the entire risk framework of the organization. If any factor or risk changes, it is immediately visible to everyone. Computers are also much better at continuously monitoring risks than people. Monitoring risks also allows your business to ensure continuity.